Ch. 11 — P01 (K) Common Random Numbers

Problem Bank | Chapter 11 | Type: K | Difficulty:

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You are comparing two inventory policies for a retail store:

- **Policy A** (baseline): reorder point $s = 10$, order-up-to $S = 50$
- **Policy B** (new): reorder point $s = 15$, order-up-to $S = 60$

Response: mean total cost per day (holding + ordering + shortage).

Use the `simdes` package `SSInventory` model (or implement your own).

Tasks

1. **Independent streams:** Run 30 replications for each policy using independent seeds. Compute the mean cost, standard deviation, and 95% CI for each. Is the difference statistically significant?
2. **CRN implementation:** Run 30 paired replications. Both policies use the same demand stream (same seed for demand generation). Compute $D_i = \text{cost}_A(i) - \text{cost}_B(i)$ for each replication i. Report $\bar{D}$, s_D, and the 95% CI for $E[D]$.
3. **Variance reduction:** Compare s_D^2 (CRN) to $s_A^2 + s_B^2$ (independent). Report the variance reduction ratio. Explain in 2 sentences why sharing the demand stream induces positive correlation.
4. **When does CRN help?** Suppose you instead compared:
   - Policy A: $s = 10$, $S = 50$
   - Policy C: $s = 10$, $S = 50$, but with a different supplier ($2\times$ longer lead time)

   Would CRN still help? Which stream(s) should be shared? Describe the synchronisation challenge when the two systems have structurally different event sequences.
5. **Effect size:** Based on the CRN paired-difference CI, is the cost difference between A and B practically significant (i.e., worth the operational change)? Define "practical significance" for this problem.